



RESPONSIVE CITIES

DESIGN MATTER(S) FOR DECARBONIZATION

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Areti Markopoulou
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INNOVATIVE COMPOSITE FORMULATIONS FOR THERMAL REGULATION:

A Synergy of Computational Design and Waste Material Valorisation to Enhance Outdoor Thermal Comfort

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KEYWORDS

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ABSTRACT

The role of computational design in developing composite materials to mitigate urban heat island (UHI) effects is the main objective of this study. Utilizing the CES GRANTA material selector, the research systematically evaluates various composite formulations that integrate waste materials, such as polyethylene terephthalate (PET), styrene-butadiene rubber (SBR) and Glass into concrete matrices. The methodology comprises four key steps: matrix material selection, particulate definition, simulation for data generation, and comparative analysis of thermal properties. The software facilitates the identification of optimal material compositions by analysing thermal conductivity, specific heat capacity, and other relevant characteristics. By running simulations based on defined parameters, the software generates updated material properties, allowing for thorough comparisons between newly designed composites and original matrix materials. Results indicate that incorporating waste materials significantly enhances the thermal efficiency of concrete, contributing to sustainability goals while addressing UHI challenges. The findings underscore the potential of computational methods in material design, enabling architects and engineers to develop innovative solutions tailored to specific environmental needs. By leveraging advanced software tools, this research demonstrates how computational approaches can transform material selection processes, ultimately promoting more sustainable urban environments and improving overall urban life quality.

1. INTRODUCTION

Urban heat islands (UHI) are a significant environmental concern in densely populated areas, characterized by elevated temperatures compared to surrounding rural regions (Fig.1). This phenomenon is primarily driven by anthropogenic activities and the extensive use of materials with high thermal mass, such as concrete and asphalt. The interplay between urban design, thermal properties of materials, and the effectiveness of composite materials in mitigating UHI effects has become a subject of increasing interest among researchers and urban planners. The UHI effect can have severe consequences on human health, energy consumption, and the overall quality of life in urban areas. Studies have shown that exposure to prolonged periods of high temperatures can lead to heat-related illnesses and mortality (Mora et al., 2017). Additionally, the increased energy demand for cooling buildings during hot periods contributes to higher greenhouse gas emissions and exacerbates the UHI effect (Santamouris, 2014). Therefore, it is crucial to develop strategies that can effectively mitigate the UHI phenomenon and create more comfortable and sustainable urban environments.

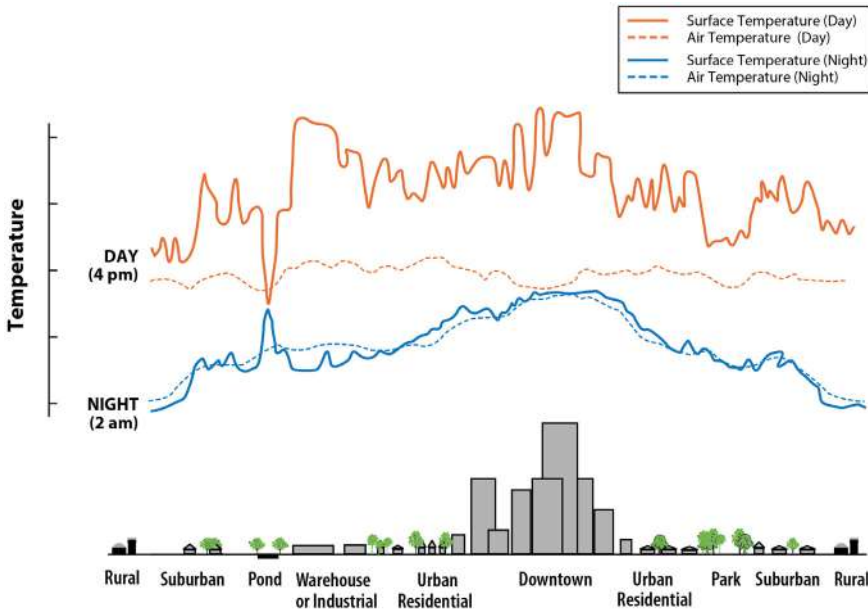


Figure 1:
Urban Heat Island (UHI)
effects and surface
temperature ("Urban Heat
Islands | U.S. Geological
Survey," 2019)

Recycling materials such as glass, PET (polyethylene terephthalate), and SBR (styrene-butadiene rubber) into new composite construction materials plays a crucial role in mitigating climate change and promoting decarbonization.

The construction sector is a significant contributor to global greenhouse gas emissions, accounting for approximately 40% of total CO2 emissions annually due to the production and transportation of building materials (Environment, 2023). Recycling PET, SBR, and glass is urgent due to environmental impact and resource conservation. These materials, used in packaging and construction, offer benefits in energy savings, emissions reduction, and waste minimization when recycled. This process reduces virgin material production and energy consumption, aligning with sustainability goals and climate change mitigation efforts. The urgency is recognized by industries and policymakers due to its significant role in environmental protection and resource management. By incorporating recycled materials into construction, the industry can reduce its carbon footprint and contribute to a more sustainable built environment. This approach aligns with global efforts to achieve net-zero emissions by mid-century, as outlined by international frameworks like the Global Alliance for Buildings and Construction (Environment, 2023).

The use of recycled glass in construction materials offers substantial environmental benefits. Research has demonstrated that recycled glass can be effectively used as a partial replacement for cement in concrete mixtures, enhancing the mechanical properties of the concrete while significantly reducing CO2 emissions (Almusaed et al., 2024). This reduction is critical given that cement production alone accounts for 8% of global CO2 emissions (Scrivener, 2024). By substituting conventional materials with recycled alternatives, the construction industry can make significant strides toward decarbonization.

Recycling PET and SBR into construction materials has shown promise in reducing environmental impacts. Gu and Ozbakkaloglu (2016) highlighted that incorporating recycled PET into concrete mixtures not only improves durability but also reduces the material's environmental footprint (Gu and Ozbakkaloglu, 2016). The use of recycled SBR in asphalt mixtures enhances their performance while decreasing reliance on virgin materials, thus contributing to decarbonization efforts (Almusaed et al., 2024). These practices underscore the potential for recycled materials to transform the construction sector into a more sustainable industry.

Decarbonization of building materials is essential for addressing climate change effectively. The UNEP report emphasizes three strategies: avoiding unnecessary material extraction, shifting to regenerative materials, and improving decarbonization of conventional materials through technological advancements (Environment, 2023). These strategies are crucial for reducing embodied carbon emissions—those associated with the production and use of building materials—which currently account for a significant portion of the sector's total emissions (United Nations Environment Programme, 2024). By adopting these approaches, the construction industry can align with global climate goals and enhance resilience against climate change impacts.

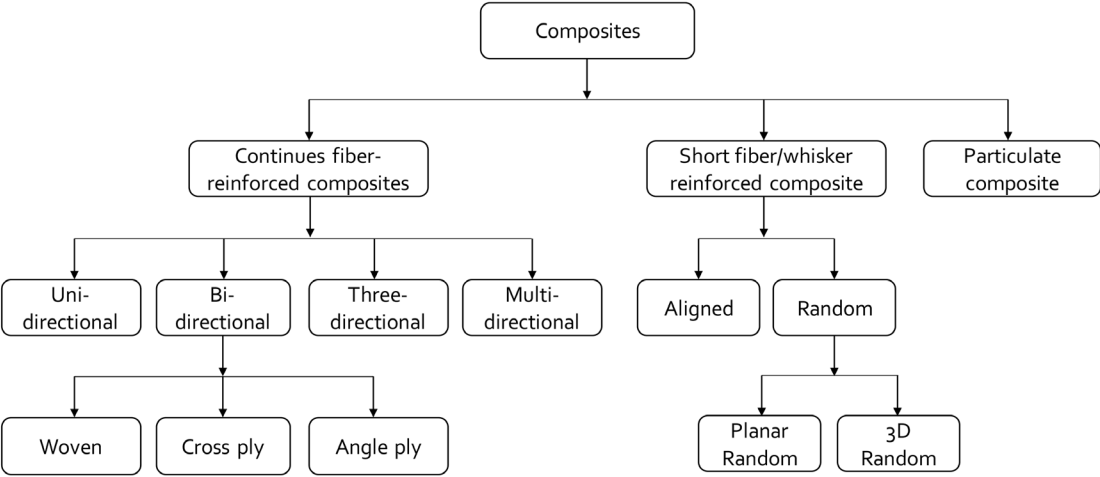


Figure 2:
Non-Renewable Materials
(Plastic & /Glass) (United
Nations Environment
Programme, 2024)

According to research conducted by the United Nations, (Fig.2) enhancing the recycling rates of plastics and polymers necessitates significant advancements in several key areas, including the collection and sorting processes, as well as the predominant reliance on mechanical recycling methods. Furthermore, there is a critical need for substantial progress in the realm of chemical recycling to effectively address the challenges associated with plastic waste. In the case of glass, the foremost priority should be the decarbonization of production processes, alongside the facilitation of recycling specifically for window glass, which remains a vital component of sustainable glass management (United Nations Environment Programme, 2024).

The term composite materials (Fig.3) refer to the integration of two or more different materials to attain improved or specific properties.

Figure 3:
Classification of composites
based on dispersed phase
(Balasubramanian, 2013)



In such composites, the dispersed phase(s) are situated within a continuous medium referred to as the matrix (Fig.4). A critical requirement is that the dispersed phase must preserve its original characteristics throughout the processing stages and during any future service or reprocessing (Balasubramanian, 2013) . in this study the focus will be on designing Particulate composite material which basically consist of matrix and particulate.

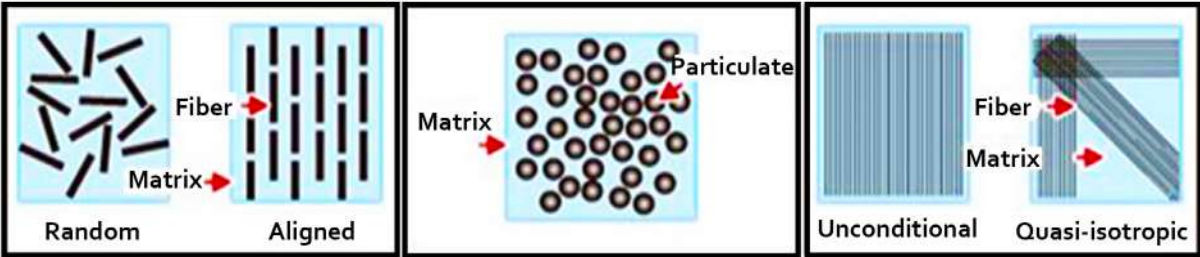


Figure 4:
Composite material
combination Base on Matrix

Composite materials, particularly those based on concrete, have the potential to enhance thermal performance and reduce heat absorption in urban canyons. The incorporation of materials such as PET or SBR into concrete composites can improve thermal properties and leading to lower surface temperatures (Irfeey et al., 2023). These composite materials not only contribute to UHI mitigation but also promote sustainability by utilizing recycled or waste materials.

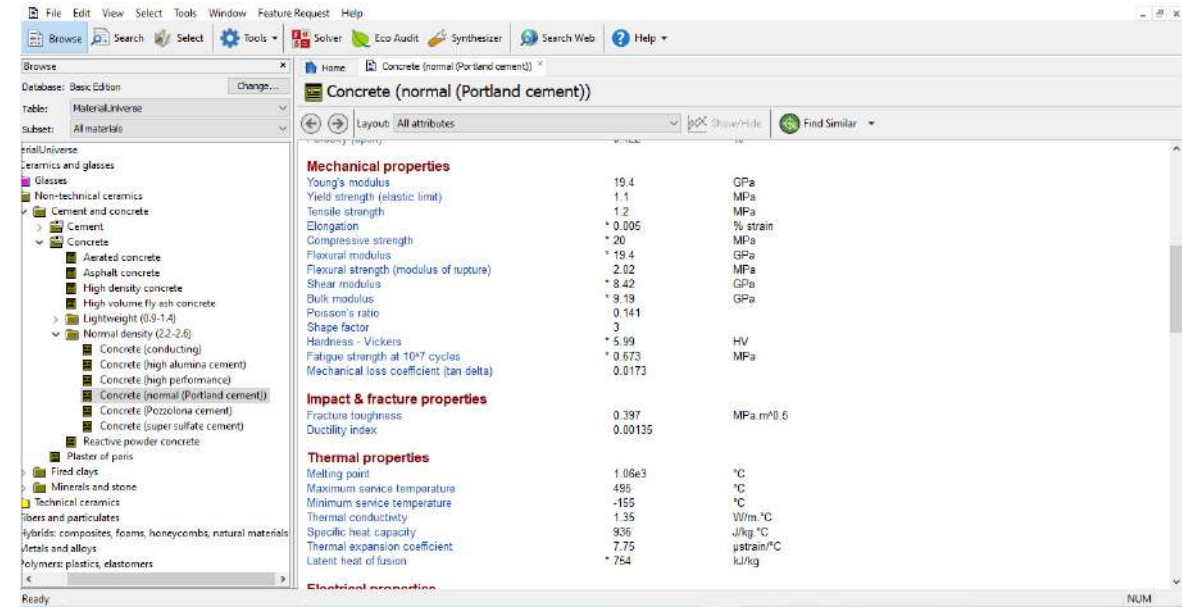


Figure 5:
CES Material selector -
material properties

The CES GRANTA material selector is a valuable tool for developing new composite materials for urban applications. It allows researchers and designers to explore material properties and compositions, helping create composites that meet specific thermal performance criteria (Fig.5). The software analyses thermal conductivity, specific heat capacity, and other relevant properties to optimize composites for urban environments.

With its extensive database and advanced algorithms, the CES GRANTA selector aids in identifying and managing materials that meet precise specifications, enhancing the design process and ensuring products align with performance requirements.

2. METHODOLOGY

The approach employed in utilizing this software facilitates the establishment of a standardized framework for both the production and application of materials. This systematic methodology not only enhances the consistency of material selection but also allows for a coherent comparison of results within a clearly delineated cycle. Such a structured process significantly contributes to the clarity and comprehensibility of the data generated, thereby improving the overall decision-making framework. The process comprises four primary steps (Table 1).

Step	Description	Key Components
1. Matrix Material Selection	Choose the base material for the composite	- Polymers - Ceramics - Other suitable materials
2. Particulate Definition and Selection	Define and select the particulate to be incorporated	- Particle size distribution - Thermal properties - Physical properties - Matrix-to-particulate ratio
3. Simulation and Data Generation	Run simulations using software to generate data	- Updated material properties - Potential behaviour alterations - Outdoor space thermal performance
4. Comparative Analysis	Compare new composites with original matrix and variations	- Newly designed composite materials - Original matrix material - Composite iterations with different particulate percentages

Table 1:
Methodological chart

By implementing this software, a methodological pathway is created that standardizes the processes involved in material production and utilization. This systematic approach enables a more effective comparison of material selection outcomes, as it operates within a well-defined cycle. Consequently, the data produced becomes more accessible and interpretable, fostering a deeper understanding of the material properties and their implications for various applications. The methodology for utilizing the software follows a systematic cycle to obtain the final data.

2.1 MATRIX MATERIAL SELECTION

The initial step involves the selection of the matrix material for the composite. This matrix can be chosen from a wide range of materials, including but not limited to polymers, ceramics, and other suitable substances (Fig.6).

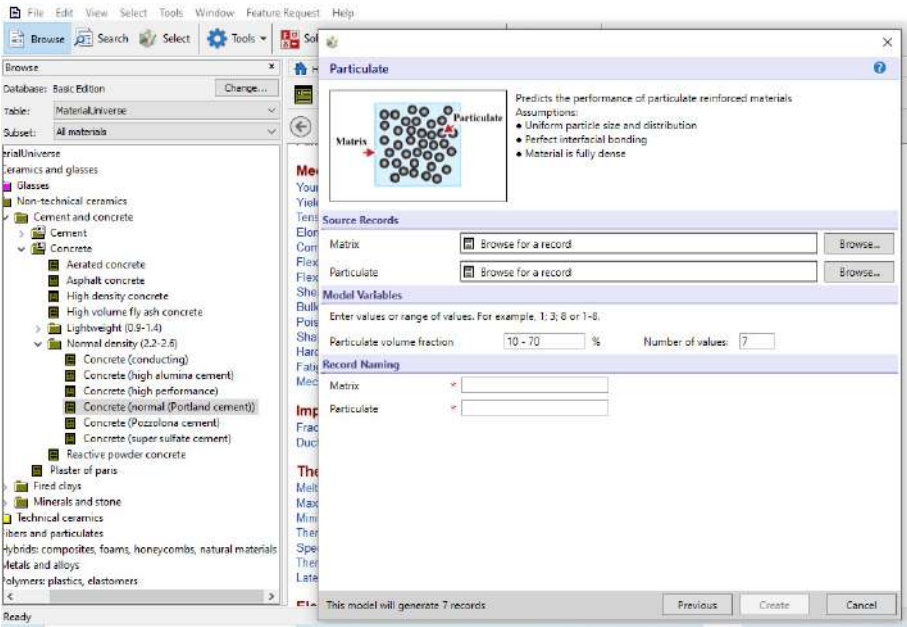


Figure 6:
CES Material selector
composite synthesizer

2.2 PARTICULATE DEFINITION AND SELECTION

The second step focuses on the selection and definition of the particulate to be incorporated into the matrix. This process requires careful consideration of several key factors:

- a) Particle size distribution
- b) Thermal properties
- c) Physical properties
- d) Matrix-to-particulate ratio (a critical parameter)

2.3 SIMULATION AND DATA GENERATION

In this phase, the software is employed to run simulations based on the defined parameters. The simulation generates data for the new composite material, including:

- a) Updated material properties, which may exhibit increases or decreases in various aspects
- b) Potential alterations in material behaviour, particularly in relation to outdoor space thermal performance

2.4 COMPARATIVE ANALYSIS

The final step involves a comprehensive comparison of:

- a) The newly designed composite materials
- b) The original matrix material
- c) Various iterations of the composite with differing particulate percentages

This comparative analysis allows for a thorough evaluation of the material's properties and performance characteristics across different compositions and return to initial step (Fig.7).

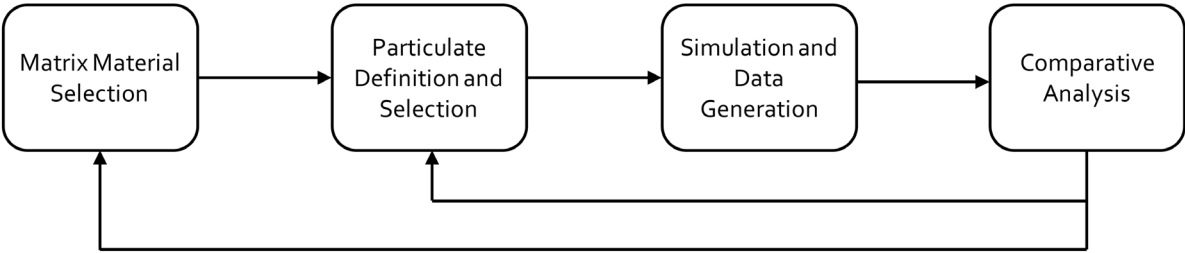


Figure 7:
Workflow Flowchart base on
CES Material Selector

3. FINDINGS AND RESULTS

This study examines a concrete matrix with Portland cement, incorporating waste materials like PET, SBR, and Glass at 10%, 30%, and 50% volumes. Using specialized software, it analyses changes in density, thermal conductivity, specific heat capacity, and thermal expansion. The focus is on how waste materials affect thermal characteristics rather than mechanical properties. The study highlights recycled materials' benefits for concrete's thermal performance in construction. Three formulations using plastic waste promote sustainability through upcycling, reducing environmental impact and CO2 emissions. It aims to improve thermal efficiency and address waste management in construction.

To verify the outcomes generated by the CES GRANTA software, we establish a specific urban space model referred to as an urban canyon, characterized by fixed dimensions of (H) = 15, (W) = 15, (L) = 60 meters. Subsequently, we implement the newly identified composite materials and conduct microclimatic simulations using ENVI-met to assess the actual impact of these materials on outdoor thermal comfort. In this investigation, the primary emphasis is placed on the materials, while the dimensions of the urban canyon remain constant to isolate the effects of the materials themselves.

The concept of an “urban canyon” refers to the vertical space created by buildings on either side of a street, which significantly influences microclimatic conditions such as sunlight exposure, wind patterns, and thermal comfort. The canyon's geometry will be designed to optimize these factors, considering the potential for heat absorption and retention, which are critical for urban heat island studies. Research has shown that the geometry of urban canyons can affect the distribution of solar radiation and the resultant thermal environment (Alcoforado and Matzarakis, 2010).

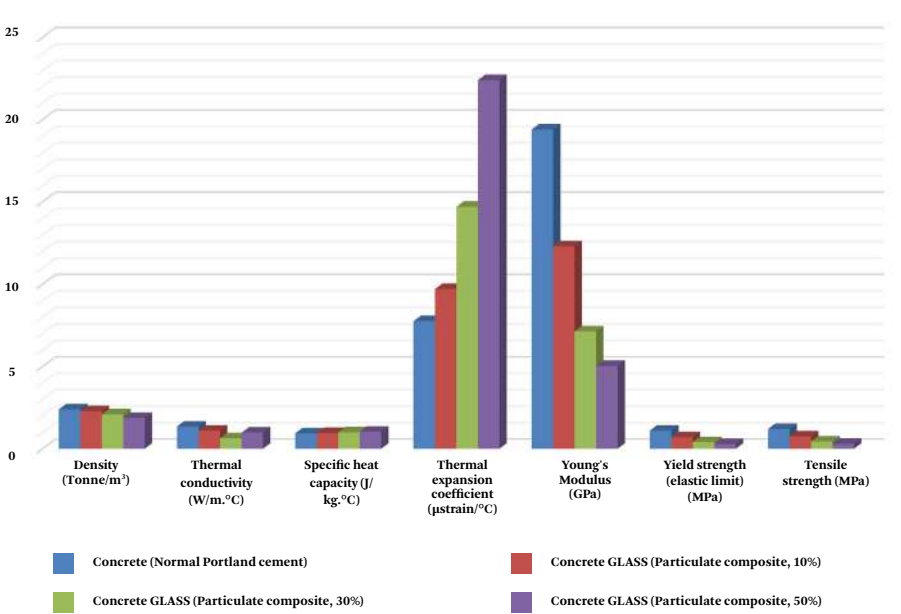
3.1. CONCRETE - PET

The initial case involves the concrete-PET composite, with results and comparisons between the original concrete and the composite mixture presented in tables 2 & 3.

	Concrete (Normal Portland cement)	Concrete PET (Particulate composite, 10%)	Concrete PET (Particulate composite, 30%)	Concrete PET (Particulate composite, 50%)
Density (kg/m³)	2390	2290	2080	1870
Thermal conductivity (W/m.°C)	1.35	1.1	0.632	0.974
Specific heat capacity (J/kg.°C)	936	952	988	1030
Thermal expansion coefficient (µstrain/°C)	7.75	9.7	14.7	22.4
Young's modulus (GPa)	19.4	12.3	7.13	5.03
Yield strength (elastic limit) (MPa)	1.1	0.694	0.403	0.284
Tensile strength (MPa)	1.2	0.756	0.439	0.31

Table 2:
Concrete – PET composite (10%, 30%, 50%) achieved properties base on software simulation

Table 3:
Concrete – PET composite (10%, 30%, 50%) properties comparison diagram



3.2. CONCRETE - SBR

The subsequent case examines the concrete-SBR composite, where tables 4 & 5 illustrates the results and distinctions between the original concrete and the composite mixture.

	Concrete (Normal Portland cement)	Concrete SBR (Particulate composite, 10%)	Concrete SBR (Particulate composite, 30%)	Concrete SBR (Particulate composite, 50%)
Density (kg/m³)	2390	2290	2020	1770
Thermal conductivity (W/m.°C)	1.35	1.22	0.957	0.682
Specific heat capacity (J/kg.°C)	936	966	1040	1130
Thermal expansion coefficient (µstrain/°C)	7.75	7.75	7.76	7.79
Young's modulus (GPa)	19.4	0.0476	0.0159	0.00955
Yield strength (elastic limit) (MPa)	1.1	0.0027	0.0009	0.00054
Tensile strength (MPa)	1.2	0.00294	0.000982	0.00059

Table 4:
Concrete – SBR composite (10%, 30%, 50%) achieved properties base on software simulation

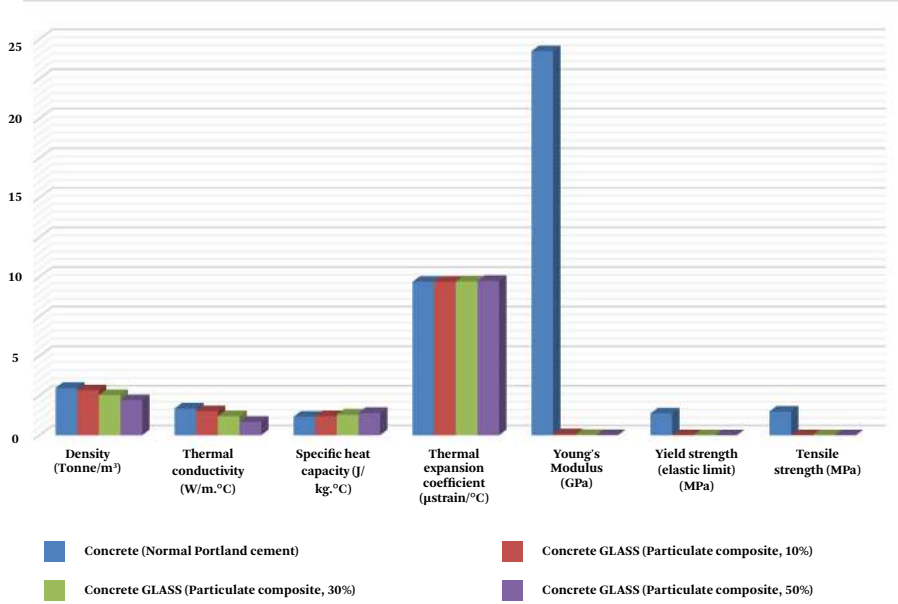


Table 5:
Concrete – SBR composite
(10%, 30%, 50%) properties
comparison diagram

3.3. CONCRETE - GLASS

The final case focuses on the concrete-Glass composite, as shown in tables 6&7, which details the results and variations between the original concrete and the composite mixture.

	Concrete (Normal Portland cement)	Concrete Glass (Particulate composite, 10%)	Concrete Glass (Particulate composite, 30%)	Concrete Glass (Particulate composite, 50%)
Density (kg/m ³)	2390	2400	2420	2440
Thermal conductivity (W/m.°C)	1.35	1.27	1.12	0.949
Specific heat capacity (J/kg.°C)	936	933	925	917
Thermal expansion coefficient (µstrain/°C)	7.75	8.29	8.73	8.92
Young's modulus (GPa)	19.4	20.9	24.7	30.2
Yield strength (elastic limit) (MPa)	1.1	1.18	1.4	1.71
Tensile strength (MPa)	1.2	1.29	1.52	1.86

Table 6:
Concrete – Glass composite
(10%, 30%, 50%) achieved
properties base on software
simulation

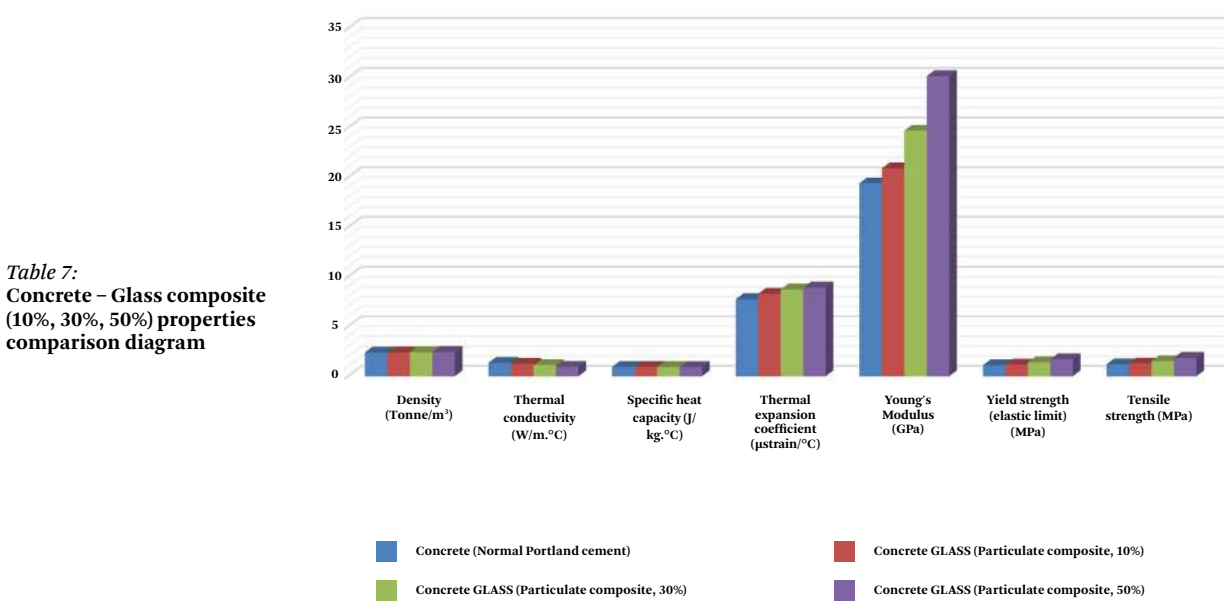
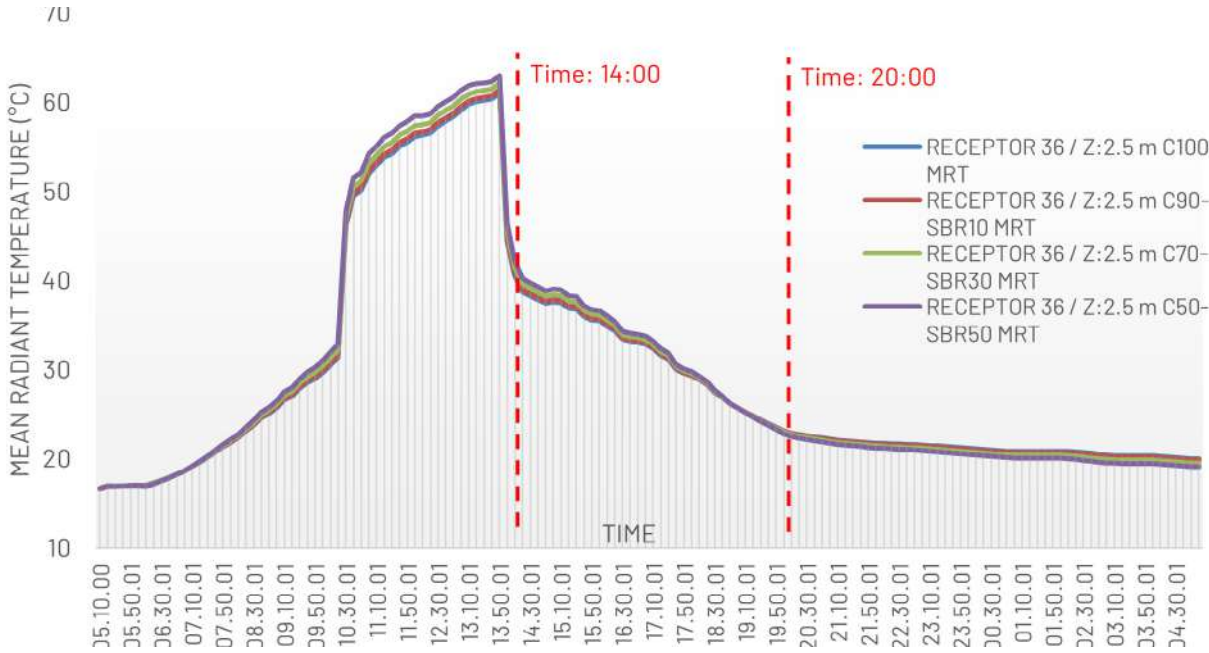


Table 7:
Concrete – Glass composite
(10%, 30%, 50%) properties
comparison diagram

A newly developed material was tested on a small urban unit, specifically an urban canyon measuring (W:15m, L:60m, H:15m), to evaluate the spatial behaviour of various extracted composites. For example, as illustrated in Figure 8 and Table 8, the Mean Radiant Temperature (MRT) for different types of concrete - SBR (50%) composite, exhibited varying performance at two distinct times on August 15th in Ancona, Italy. At 14:00, the results indicated a higher temperature than the original matrix, while by 20:00, the canyon experienced a cooler temperature.

Table 8:
Concrete – SBR (10%,
30%, 50%) Mean Radiant
Temperature



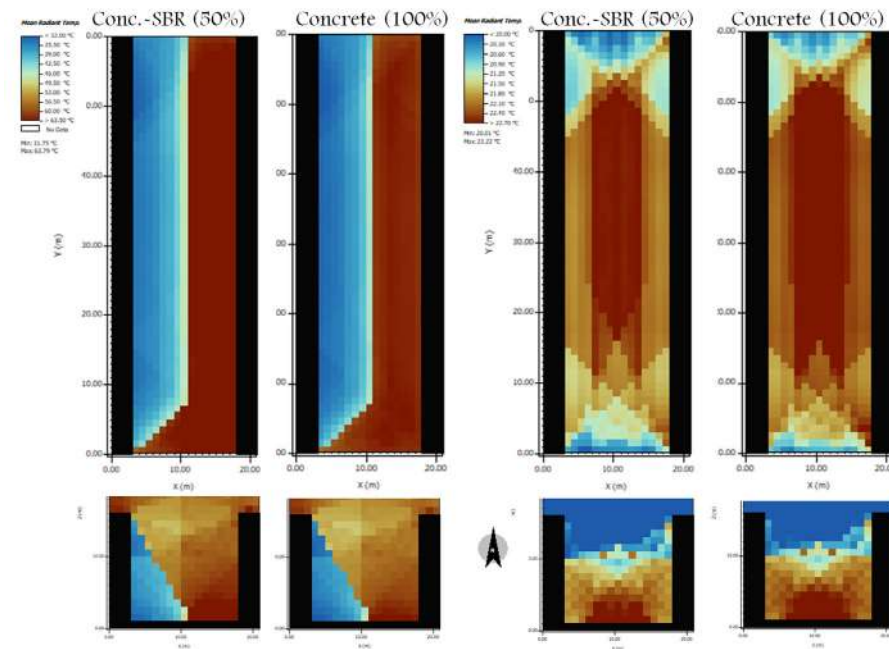


Figure 8:
Concrete (100%) - Concrete
- SBR (50%) Mean Radiant
Temperature map at 14:00
(Right), 20:00 (Left)

This behaviour aligns with predictions based on the enhanced properties achieved through computational material design. The results of the outdoor simulation are illustrated in Figures 9 and 10. These graphics display the outcomes of various composite mixtures analysed by the software applied to the canyon in ENVI-met, specifically at 14:00 and 20:00. The comparison reveals a notable distinction between the results of pure concrete and those of the 50% extracted composite.

Figure 9:
Different composites (50%)
Mean Radiant Temperature
map at 14:00

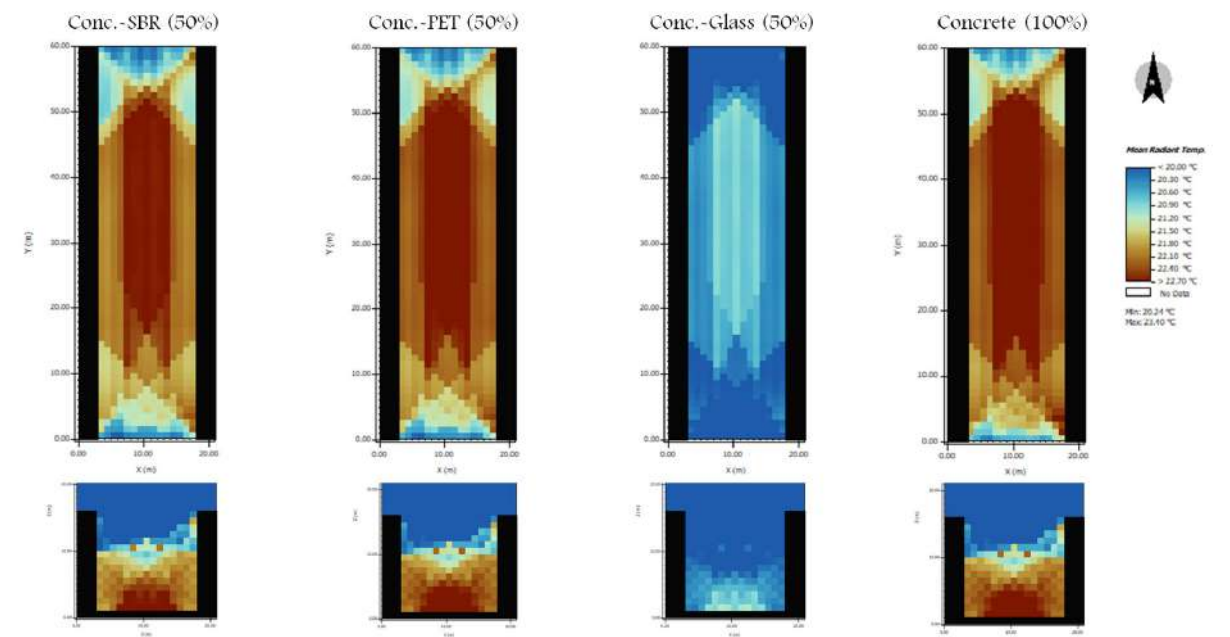
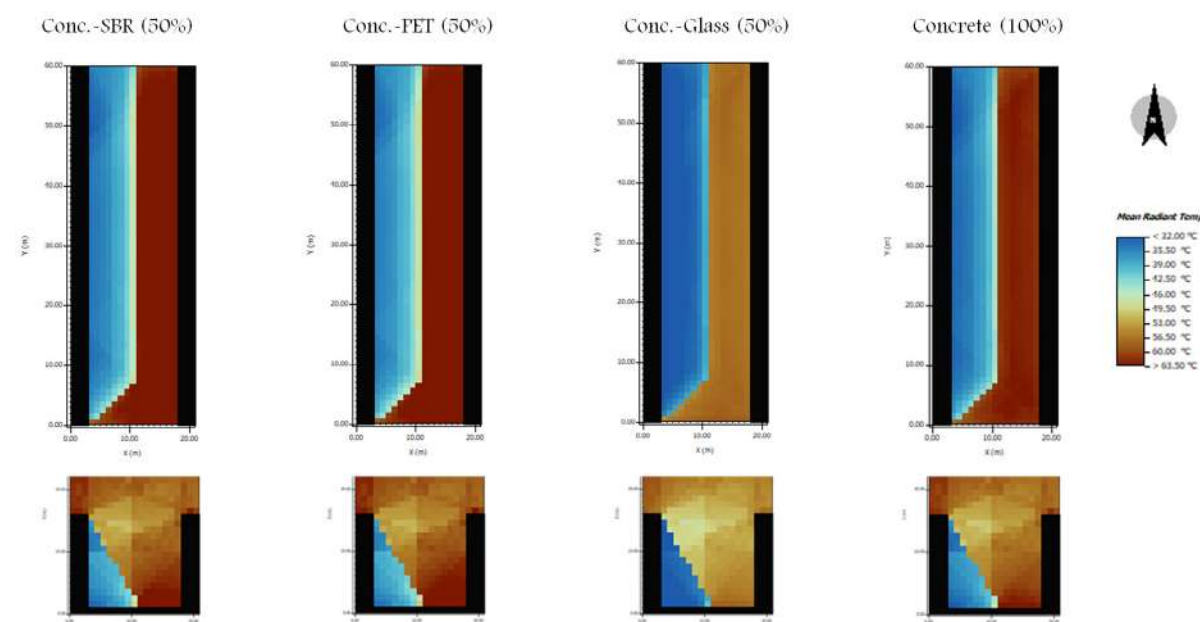


Figure 10:
Different composites (50%)
Mean Radiant Temperature
map at 20:00

In Figure 9, which illustrates the comparison of the results between the composites and pure concrete at 14:00, it is evident that while the concrete-glass composite shows an enhancement, the concrete-PET and concrete-SBR composites exhibit poorer performance. These outcomes can be attributed to the distinct material properties, particularly the thermal characteristics, that were obtained during the composite material extraction process in CES.

Figure 10 further illustrates the comparative performance of four different composites, each with a 50% mix, at 20:00, revealing that polymer composites perform slightly better than the original matrix, with glass composites showing significant improvement. The findings indicate the significant role of materials and composite materials in addressing urban heat islands and enhancing outdoor thermal comfort.

4. CONCLUSION AND FURTHER RESEARCH OPPORTUNITIES

The analysis of the results and graphical representations indicates that the performance of concrete composites varies significantly over time.

- At 14:00, the glass composite demonstrates superior efficacy in reducing the Mean Radiant Temperature (MRT), thereby enhancing thermal comfort. In contrast, the concrete composites incorporating PET and SBR at a 50% ratio exhibit less favourable outcomes, as they contribute to an increase in temperature rather than a decrease.

- By 20:00, all three types of composites reveal improved outdoor conditions, which collectively foster a greater sense of thermal comfort within urban environments. This shift suggests that, despite the initial performance discrepancies observed earlier in the day, the composites ultimately contribute positively to the thermal dynamics of the space, enhancing the overall comfort levels for individuals in the area.

The methodology and results underscore the critical significance of material selection and the introduction of innovative materials aimed at alleviating the urban heat island effect and enhancing outdoor thermal comfort. The utilization of CES GRANTA provides substantial opportunities for the digital study and reproduction of materials, circumventing many of the constraints associated with traditional material production and laboratory testing. This digital approach facilitates a comprehensive analysis of material performance prior to their physical manifestation and application, thereby allowing for a more informed selection process.

This approach aligns with the idea of digitizing physical materials, facilitating the anticipation of material performance before real-world application. It establishes a unique identification system or "passport" for materials, deepening our understanding of their characteristics. Although this study did not include life cycle assessment (LCA), future research could explore this. This framework supports efficient material utilization and promotes a strategic methodology in urban planning, enhancing environmental quality and human well-being.

Specialized software analysis enables architects, engineers, and designers to select appropriate materials for projects, addressing urban heat islands and environmental challenges. This process enhances designs to meet sustainability requirements in urban development. Computer-aided material design methodologies support the reuse of polymer-based waste materials, which are often difficult to recycle chemically and produce significant carbon emissions. This approach promotes a circular economy in construction and design sectors. By focusing on waste material valorisation and leveraging their unique properties, such as thermal characteristics, computer-aided design facilitates greener solutions. This methodology represents a shift towards more sustainable practices in architecture and engineering, fostering environmentally friendly structures and mitigating adverse urban effects.

While the findings presented are primarily derived from digital simulations and computer-based methodologies, there exists a significant opportunity to extend this approach into practical applications. By fabricating materials informed by the digital data, researchers can conduct empirical tests in laboratory settings to verify the results.

This validation process is particularly crucial for assessing the properties of new materials, which are essential for developing effective mitigation strategies and enhancing the overall performance of materials in both building and urban-scale analyses.

The transition from digital simulation to real-world application not only allows for the confirmation of theoretical results but also facilitates a deeper understanding of material behaviour under practical conditions. Such empirical investigations are vital for ensuring that the innovative materials developed through computational methods meet the necessary performance standards in real-life scenarios. Ultimately, this integration of digital and physical testing will contribute to more robust and resilient building practices, thereby improving urban infrastructure and sustainability.

REFERENCES

Alcoforado, M.-J., Matzarakis, A., 2010. Planning with urban climate in different climate zones. *Geographica* 57, 5–39. https://doi.org/10.26754/ojs_geoph/geoph.201057808

Almusaed, A., Yitmen, I., Myhren, J.A., Almssad, A., 2024. Assessing the Impact of Recycled Building Materials on Environmental Sustainability and Energy Efficiency: A Comprehensive Framework for Reducing Greenhouse Gas Emissions. *Buildings* 14, 1566. <https://doi.org/10.3390/buildings14061566>

Balasubramanian, M., 2013. *Composite Materials and Processing*. CRC Press, Boca Raton. <https://doi.org/10.1201/b15551>

Environment, U.N., 2023. *Building Materials And The Climate: Constructing A New Future | UNEP - UN Environment Programme* [WWW Document]. URL <https://www.unep.org/resources/report/building-materials-and-climate-constructing-new-future> (accessed 11.25.24).

Gu, L., Ozbakkaloglu, T., 2016. Use of recycled plastics in concrete: A critical review. *Waste Manag.* 51, 19–42. <https://doi.org/10.1016/j.wasman.2016.03.005>

Irfeey, A.M.M., Chau, H.-W., Sumaiya, M.M.F., Wai, C.Y., Muttill, N., Jamei, E., 2023. Sustainable Mitigation Strategies for Urban Heat Island Effects in Urban Areas. *Sustainability* 15, 10767. <https://doi.org/10.3390/su151410767>

Mora, C., Dousset, B., Caldwell, I.R., Powell, F.E., Geronimo, R.C., Bielecki, C.R., Counsell, C.W.W., Dietrich, B.S., Johnston, E.T., Louis, L.V., Lucas, M.P., McKenzie, M.M., Shea, A.G., Tseng, H., Giambelluca, T.W., Leon, L.R., Hawkins, E., Trauernicht, C., 2017. Global risk of deadly heat. *Nat. Clim. Change* 7, 501–506. <https://doi.org/10.1038/nclimate3322>

Santamouris, M., 2014. Cooling the cities – A review of reflective and green roof mitigation technologies to fight heat island and improve comfort in urban environments. *Sol. Energy* 103, 682–703. <https://doi.org/10.1016/j.solener.2012.07.003>

Scrivener, K., 2024. *Towards a Strategic Roadmap for Decarbonizing Building Materials*.

United Nations Environment Programme, 2024. *2023 Global Status Report for Buildings and Construction: Beyond foundations - Mainstreaming sustainable solutions to cut emissions from the buildings sector*. United Nations Environment Programme. <https://doi.org/10.59117/20.500.11822/45095>

Urban Heat Islands | U.S. Geological Survey [WWW Document], 2019. URL <https://www.usgs.gov/media/images/urban-heat-islands> (accessed 10.14.24).



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